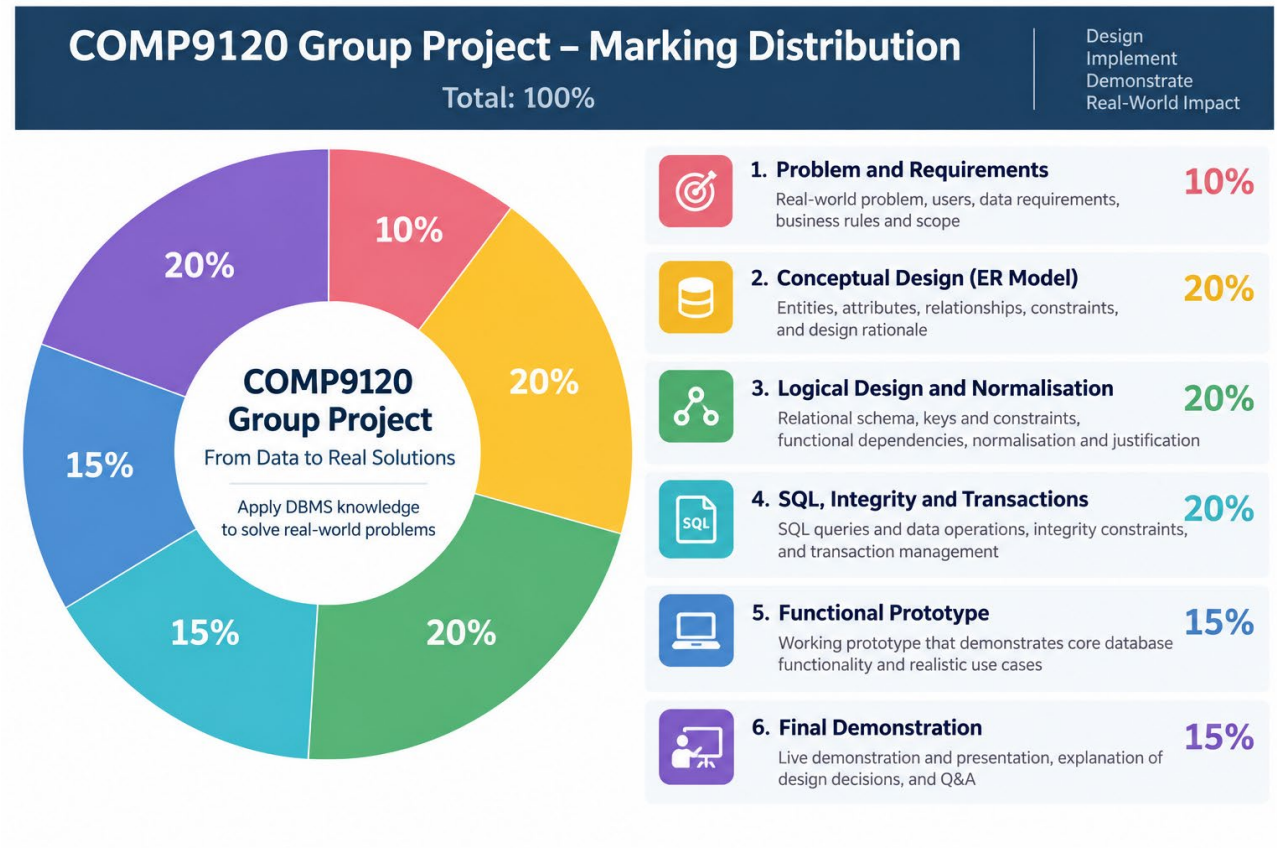
Group Project Rubric

Real-World Database Prototype

This rubric uses grade-band standards. Markers make a holistic judgement within each weighted criterion. The descriptors are not additive checkboxes. The report must contain explicit sections for criteria #1-#4 so that assessment evidence is easy to locate.

1. Marking Distribution



Criterion	Weight	Primary evidence
1. Problem and Requirements	10%	Report section #1
2. Conceptual Design (ER Model)	20%	Report section #2
3. Logical Design and Normalisation	20%	Report section #3
4. SQL, Integrity and Transactions	20%	Report section #4 + submitted implementation
5. Functional Prototype	15%	Working prototype + max 3-minute showcase video
6. Final Demonstration	15%	One-page poster + Week 12/13 live poster showcase, prototype demonstration and Q&A

2. Grade-Band Interpretation

Pass (50-64)

Core requirements are present and mostly correct. The project demonstrates acceptable achievement of the relevant learning outcomes, although reasoning, implementation or communication may be basic.

Credit (65-74)

The solution is coherent and functional, with sound application of database concepts and clear justification of important decisions.

Distinction (75-84)

The project demonstrates strong technical reasoning, thoughtful design choices, robust implementation and convincing integration of the relevant concepts.

High Distinction (85-100)

The work demonstrates sophisticated, coherent and professional application of the material available by Week 10, with strong evidence, precision, creativity and critical judgement.

3. Detailed Rubric

1. Problem and Requirements - 10%

Band	Descriptor
Pass 50-64	A suitable real-world database problem is identified. Basic users, data requirements, business rules and operations are stated, though some may be vague.
Credit 65-74	The problem and scope are clear. Users, requirements, business rules and representative operations are coherent and sufficiently detailed to guide design.
Distinction 75-84	Requirements are precise, realistic and traceable to the database design. Important assumptions, edge cases and rules are thoughtfully identified.
High Distinction 85-100	Problem framing is exceptionally clear and compelling. Requirements demonstrate sophisticated domain reasoning and form a strong, internally consistent foundation for the solution.

2. Conceptual Design (ER Model) - 20%

Band	Descriptor
Pass 50-64	An ER model represents the main entities and relationships. Most attributes, cardinalities and participation constraints are plausible, with limited modelling errors.
Credit 65-74	The ER model accurately captures important requirements using appropriate entities, attributes, relationships and constraints. Important choices are explained.
Distinction 75-84	The conceptual model handles non-trivial domain structure elegantly. Constraints and modelling

	decisions are precise, consistent and strongly justified.
High Distinction 85-100	The model is exceptionally coherent, expressive and economical, capturing subtle requirements without unnecessary complexity and demonstrating mastery of conceptual modelling.

3. Logical Design and Normalisation - 20%

Band	Descriptor
Pass 50-64	The ER model is translated into a workable relational schema. Primary/foreign keys are mostly correct. Basic functional dependencies and normalisation are addressed.
Credit 65-74	The relational design is sound and traceable to the conceptual model. Keys and constraints are appropriate; functional dependencies and normalisation decisions are correctly analysed.
Distinction 75-84	The schema is robust and carefully refined. Normalisation reasoning is technically strong and design trade-offs are explicitly justified.
High Distinction 85-100	The logical design demonstrates exceptional precision and insight. Functional dependencies, decomposition and design trade-offs are rigorously analysed and produce an elegant schema.

4. SQL, Integrity and Transactions - 20%

Band	Descriptor
Pass 50-64	The PostgreSQL database can be created and populated. Core SQL operations work; essential integrity constraints are present; at least one meaningful multi-step transaction is identified and basic failure behaviour is explained.
Credit 65-74	SQL reliably supports the stated requirements and uses an appropriate range of techniques. Constraints enforce important rules. A realistic transaction is correctly implemented/demonstrated with sound reasoning about correctness and rollback.
Distinction 75-84	SQL is technically strong and well matched to realistic requirements. Integrity is enforced thoughtfully. Transaction boundaries, failure behaviour and relevant concurrency issues are carefully designed and convincingly explained.
High Distinction 85-100	The implementation is exceptionally coherent and professional. SQL is precise and expressive, integrity mechanisms are comprehensive, and transaction design demonstrates rigorous reasoning about correctness, failure and relevant concurrency within the Week 10 scope.

5. Functional Prototype - 15%

Band	Descriptor
Pass 50-64	A working prototype connects to PostgreSQL and

	demonstrates the main database operations. The submitted video shows core features within the 3-minute limit, though communication or usability may be basic.
Credit 65-74	The prototype reliably demonstrates major workflows and most proposed functions work as intended. The video is clear, concise and engaging, showing how the database supports the real-world problem.
Distinction 75-84	The prototype is robust and well integrated with the database. The video communicates the problem, solution and user experience effectively and demonstrates thoughtful creativity in the project showcase.
High Distinction 85-100	The prototype provides an exceptionally convincing end-to-end experience. The showcase video is polished, highly effective and creatively communicates the project while clearly demonstrating authentic database functionality.

6. Final Demonstration - 15%

Band	Descriptor
Pass 50-64	The group provides the required one-page poster, participates in the showcase, demonstrates the prototype and can answer basic questions about the project.
Credit 65-74	The poster communicates the project clearly. The live demonstration is organised and convincing, and group members can explain important database decisions and respond appropriately to questions.
Distinction 75-84	The poster is polished and visually effective. The demonstration runs smoothly, connects the real-world problem to the database solution, and group members confidently defend technical choices during discussion and Q&A.
High Distinction 85-100	The poster and live showcase are at an exceptional professional standard. The group communicates a compelling project story, demonstrates the prototype confidently, engages the audience effectively and shows mastery when discussing and defending the database solution.

4. Submission and Evidence Requirements

- One-page contribution summary using the teaching-team template.
- Report with four explicit main sections corresponding exactly to marking criteria #1-#4. Evidence placed elsewhere may be harder for markers to locate; students should make the alignment explicit.
- Maximum 3-minute functional prototype showcase video. Creativity is encouraged; the most creative showcase video will receive a Certificate of Appreciation from the teaching team.
- One-page poster submitted in Week 11 and later printed by the group for the Week 12 or 13 poster showcase event.
- Required technical scope is limited to content taught by the end of Week 10.

5. Holistic Marking Guidance

A project does not need to satisfy every phrase in a higher-band descriptor to receive a mark within that band. Markers should consider the overall quality and consistency of evidence for each weighted criterion. Strong presentation cannot compensate for serious technical errors in the corresponding database criteria. The Certificate of Appreciation for the most creative showcase video is recognition from the teaching team and does not add marks beyond the Functional Prototype criterion.